

# Cristobal Cortés Gómez

Software Developer & Mathematician

legocris@hotmail.com | +52 33 1115 0013 | Guadalajara, Mexico — open to relocation  
Portfolio: [legocris.github.io](https://legocris.github.io) | LinkedIn: [www.linkedin.com/in/cristobal-cortes-gomez](https://www.linkedin.com/in/cristobal-cortes-gomez)

## Professional Profile

Cross-platform developer across game, web and embedded systems — mostly game development, with the most recent projects built on web technologies. Experienced in porting, extending and debugging existing codebases. Mathematics graduate with a background in machine learning; advanced English, used daily with U.S.-based clients.

## Technical Skills

**Programming languages:** TypeScript/JavaScript, Python, C, C#, GML, GLSL.

**Web, frameworks & tools:** HTML/CSS, Node.js, React, Redux Toolkit, Three.js, Firebase, GameMaker Studio, Unity 3D.

**Systems & hardware:** Linux (shell scripting), Git, Emacs, Vim; ESP8266, Arduino, PLC.

**Spoken languages:** Spanish (native); English (advanced).

## Experience

**2025 – Jun. 2026:** Software Developer — *Gnarled Helix LLC (U.S. studio), remote contract*

- Developed a 2D action game that teaches programming (TypeScript, Three.js, React, Redux, Node.js), built around a runtime that executes player-written JavaScript to drive the character's abilities.
- Worked within a multidisciplinary, distributed team on an existing codebase.

**2024–2025:** Designer & Sole Developer — *Jalapeño Labs, contract*

- Led the video game adaptation of the board game “Camino a Xibalbá PITZ” (GameMaker Studio, JavaScript, Firebase), translating a printed rule set into working game logic and checking behavior against those rules case by case.
- Owned the correctness of turn and rule resolution, and the Firebase data layer behind player accounts and match records.

**2023–2024:** Embedded Systems & Automation — *Private client, contract*

Built an automated greenhouse for edible mushroom cultivation, with humidity and temperature control: firmware, PCB design and the complete electrical installation (ESP8266, Arduino IoT).

**2022–2024:** Mathematics Tutor — *Independent*

**2022:** Robotics Instructor — *American World Languages*

**2020–2022:** Hardware & Software Developer — *Proyecto INTERSEX*

Built an LED face mask that displays custom images, and a 3D museum simulator, for a social and artistic project (JavaScript, Arduino, ESP8266, Blender, Unity 3D).

**2017–2020:** Web Developer — *SGT Total*

Built a mobile-responsive commercial website over several months, then handled occasional change requests from the client in the years that followed.

**2017–2018:** Freelance projects — *Upwork and direct clients*

*Immertec* (2018, Upwork): ported a GLSL shader to HLSL, delivering a spherical/fisheye VR projection for a medical application. *Daniel Douglas* (2018, Upwork): desktop-simulation mini-game built to a written client brief. “*Habitar*” theater production (2017): Kinect application adjusting real-time projections to physical set movement (Processing, Java, GLSL).

**2016–2017:** Industrial Systems & Automation — *Cervejal*

Built a semi-automatic bottle labeling machine and programmed a Siemens PLC (ladder logic) for a keg washer/filler, commissioned and tested on site.

## Selected Independent Projects

**Shipped games:** “Morbus” (2012), “Sweet Valentine’s Day” (2014), “Sacrifice Your Brothers” (2018), “Me Canso Ganso” (2019).

**Other game projects:** “Gravity Box” (2018, unreleased); “Libélula” (in development).

## Education

**2025:** Bachelor’s Degree in Mathematics — Universidad de Guadalajara (CUCEI). Final project: “Suitability Analysis of Shallow vs. Deep Neural Networks Using the Perceptron for Classification Problems”.